

SAULT SAINTE MARIE, MI, USA

WMO: 727340

Lat: 46.479N

Lon: 84.357W

Elev: 220

StdP: 98.71

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-20.5	-28.2	0.3	-23.0	-25.1	0.4	-19.7	10.7	-5.6	9.3	-6.1	2.3	80	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.7	20.9	27.1	19.9	25.5	18.9	22.3	26.8	21.2	25.1	20.2	23.9	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.9	24.7	19.8	14.9	23.4	18.8	14.0	22.3	66.4	26.7	62.5	25.1	58.9	24.1	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.2	7.4	-27.2	31.4	3.7	1.6	-29.8	32.6	-32.0	33.5	-34.1	34.5	-36.8	35.6		
			-27.4	24.4	3.7	1.2	-30.0	25.3	-32.2	26.0	-34.2	26.6	-36.9	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-9.1	-8.0	-3.1	3.8	10.9	16.1	18.8	18.5	14.8	8.0	1.4	-4.7
	DBStd	10.96	6.57	5.99	5.92	4.55	4.38	3.68	3.23	2.93	4.07	4.25	4.82	5.65
	HDD10.0	2552	593	504	406	191	42	2	0	0	7	93	259	456
	HDD18.3	4740	851	737	663	436	234	86	34	35	121	322	508	714
	CDD10.0	975	0	0	1	5	69	184	273	262	149	30	2	0
	CDD18.3	122	0	0	0	0	3	18	48	38	13	1	0	0
	CDH23.3	923	0	0	1	2	45	179	381	238	75	2	0	0
	CDH26.7	181	0	0	0	0	7	34	86	41	12	0	0	0
Wind	WSAvg	3.3	3.4	3.5	3.5	3.8	3.6	3.0	2.9	2.9	3.4	3.6	3.5	
Precipitation	PrecAvg	863	58	40	51	62	68	74	74	86	95	94	88	72
	PrecMax	1240	115	95	126	140	188	187	184	241	211	194	196	159
	PrecMin	569	13	5	4	22	16	9	14	26	13	4	22	15
	PrecStd	143	24	21	27	26	34	36	36	38	45	43	32	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	6.4	14.7	20.3	27.6	29.8	31.2	29.9	28.4	22.5	15.4	8.0
		MCWB	3.7	3.8	9.0	11.6	18.7	21.7	22.2	21.8	21.4	17.2	11.7	6.6
	2%	DB	2.9	3.5	9.8	16.7	24.2	27.4	29.0	27.7	25.3	19.2	11.7	5.3
		MCWB	1.9	1.8	5.6	9.5	16.1	19.8	21.0	20.6	19.4	15.2	9.2	4.1
	5%	DB	1.6	1.9	6.9	13.6	21.6	25.3	27.2	26.0	23.4	16.7	9.6	3.6
		MCWB	0.6	0.5	3.9	7.4	14.4	18.1	19.9	19.6	18.4	13.7	7.6	2.6
	10%	DB	0.1	0.4	4.4	10.9	18.9	23.5	25.5	24.3	21.4	14.4	7.9	2.4
		MCWB	-0.8	-0.8	2.0	6.1	12.7	17.2	19.1	18.8	17.3	11.9	6.2	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.3	10.4	13.4	20.1	23.3	24.0	23.4	22.1	18.5	12.2	7.1
		MCDB	4.6	5.6	12.9	18.8	25.7	28.4	28.7	27.3	26.4	20.3	14.3	7.9
	2%	WB	2.0	1.9	6.6	10.4	17.6	21.0	22.4	21.9	20.6	15.9	9.8	4.2
		MCDB	2.7	3.2	9.3	14.5	22.4	25.6	27.1	25.7	23.9	18.8	11.6	5.0
	5%	WB	0.7	0.7	4.1	8.3	15.5	19.6	21.2	20.8	19.4	14.0	7.9	2.8
		MCDB	1.5	1.7	6.4	12.7	19.7	23.8	25.3	24.3	22.3	16.3	9.3	3.5
	10%	WB	-0.8	-0.8	2.2	6.6	13.6	18.2	20.1	19.8	18.0	12.2	6.5	1.5
		MCDB	0.2	0.4	4.2	10.5	17.9	22.0	23.9	23.2	20.7	14.3	7.9	2.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.3	8.7	9.0	11.0	11.3	10.7	9.9	9.5	7.6	6.2	6.4
		MCDBR	6.3	7.9	10.4	14.3	15.4	14.2	13.3	12.1	11.6	10.7	8.5	6.4
		MCWBR	5.9	6.8	7.1	8.5	8.6	7.8	6.7	6.2	6.3	7.2	6.7	5.9
	5% WB	MCDBR	6.2	7.6	9.2	12.6	13.5	12.1	11.2	10.0	10.3	9.6	7.8	6.0
		MCWBR	6.0	6.9	6.8	8.4	8.5	7.5	6.6	5.9	6.4	7.0	6.7	5.8
Clear-Sky Solar Irradiance	taub	0.278	0.289	0.318	0.355	0.368	0.386	0.387	0.377	0.363	0.329	0.301	0.285	
	taud	2.336	2.293	2.328	2.331	2.376	2.347	2.362	2.388	2.426	2.507	2.508	2.367	
	Ebn at Noon	819	884	908	901	898	880	873	868	850	834	797	766	
	Edh at Noon	74	95	107	118	117	121	118	110	97	77	64	65	
All-Sky Solar Radiation	RadAvg	1.09	1.99	3.28	4.46	5.55	5.98	6.00	5.07	3.80	2.22	1.17	0.85	
	RadStd	0.12	0.22	0.37	0.50	0.46	0.41	0.34	0.33	0.29	0.25	0.15	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page